Tree-HyperLISTA: Ultra-Lightweight Deep Unfolding for Tree-Sparse Recovery

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Abstract

Tree-structured sparsity arises naturally in wavelet decompositions and hierarchical signal representations, where the support of a sparse signal respects a rooted tree topology. Existing recovery methods either use classical iterative algorithms with no learning, or deep unfolding networks with millions of parameters that ignore tree structure. We propose **Tree-HyperLISTA**, an ultra-lightweight deep unfolding network for tree-sparse recovery that uses only **3 learned hyperparameters**. Our method replaces elementwise thresholding with a tree-consistent support projection operator that combines bottom-up subtree scoring with greedy tree-consistent support selection and soft-thresholding within the selected support. On synthetic tree-sparse recovery problems (n=255, m=127), Tree-HyperLISTA achieves -29.4 dB NMSE with perfect support recall, outperforming elementwise HyperLISTA (-14.1 dB, 3 params) by over 15 dB and LISTA (-15.4 dB, 1.56M params) by 14 dB. Our results demonstrate that the "minimal parameterization" principle extends from block sparsity to hierarchical tree sparsity: strong structural inductive bias in the proximal operator eliminates the need for heavy parameterization.

1 Introduction

Sparse recovery—reconstructing a high-dimensional signal $\mathbf{x} \in \mathbb{R}^n$ from compressed measurements $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\varepsilon}$—is a foundational problem in compressed sensing **??**. Classical methods like ISTA and FISTA solve this via iterative proximal gradient descent, while deep unfolding approaches (LISTA **?**, ALISTA Liu et al. [2019], HyperLISTA Chen et al. [2021]) learn parameters of the unfolded iterations to accelerate convergence.

A key insight from recent work is that structured sparsity models—where the support of $\mathbf{x}$ obeys geometric constraints—can be exploited to dramatically improve recovery. Block sparsity, where nonzero coefficients cluster in contiguous groups, has been studied extensively Eldar et al. [2010], **?**. However, many real-world signals exhibit *tree-structured* sparsity: in wavelet decompositions of natural images, if a fine-scale coefficient is significant, its parent (coarser-scale) coefficient is typically also significant Baraniuk et al. [2010], **?**.

Formally, the coordinates of $\mathbf{x}$ are arranged in a rooted tree $\mathcal{T}$, and the support $S = \mathrm{supp}(\mathbf{x})$ satisfies *tree consistency*: $i \in S \Rightarrow \mathrm{parent}(i) \in S$ for every non-root node. Classical model-based compressed sensing Baraniuk et al. [2010] exploits this via tree-projected gradient descent, but uses no learning. Deep unfolding methods learn iteration parameters but operate elementwise, ignoring tree structure entirely.

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We bridge this gap with **Tree-HyperLISTA**, which integrates tree-structured support selection into an ultra-lightweight deep unfolding backbone with only 3 global hyperparameters (c_1, c_2, c_3). Our contributions are:

- A novel *tree-consistent support projection operator* $\mathcal{T}_{\theta,K}$ that combines subtree-aware scoring, greedy top-K tree-consistent support selection, and soft-thresholding within the selected support.
- Demonstration that 3 hyperparameters suffice for tree-sparse recovery, extending the minimal parameterization principle from block sparsity to hierarchical sparsity.
- Comprehensive experiments showing Tree-HyperLISTA outperforms all baselines by 14–24 dB NMSE with perfect support recall and only 3 parameters.

2 Related Work

Deep unfolding for sparse recovery. LISTA **?** unfolds ISTA iterations into a neural network with learned weight matrices. ALISTA Liu et al. [2019] reduces parameters by computing weights analytically, learning only per-layer step sizes and thresholds. HyperLISTA Chen et al. [2021] further reduces to 3 global hyperparameters with adaptive per-layer formulas. All these methods use elementwise thresholding and are unaware of structured sparsity.

Model-based compressed sensing. Baraniuk et al. Baraniuk et al. [2010] introduced model-based CoSaMP and model-based IHT for structured sparsity models including tree sparsity. He and Carin **?** developed tree-structured sparse Bayesian learning. These methods exploit tree structure but have no learned parameters and converge slowly.

Structured deep unfolding. Recent work has explored block-aware unfolding (Ada-BlockLISTA **?**, Struct-HyperLISTA), but tree-aware deep unfolding has not been explored. Our work is the first to combine tree-structured support selection with ultra-lightweight deep unfolding.

3 Problem Setup

We observe $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\varepsilon}$ where $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$ is tree-sparse, and $\boldsymbol{\varepsilon}$ is noise. The coordinates of $\mathbf{x}$ are arranged in a rooted tree $\mathcal{T}$ (e.g., a balanced binary tree of depth D), and the support satisfies tree consistency.

Definition 1 (Tree-sparse signal)**.** *A signal* $\mathbf{x}$ *is* K*-tree-sparse on tree* $\mathcal{T}$ *if* $|\mathrm{supp}(\mathbf{x})| \le K$ *and* $i \in \mathrm{supp}(\mathbf{x}) \Rightarrow \mathrm{parent}(i) \in \mathrm{supp}(\mathbf{x})$ *for all non-root* i*.*

We assume $\mathbf{A}$ satisfies the tree-RIP Baraniuk et al. [2010]: for all K-tree-sparse $\mathbf{x}$, $(1-\delta_K)\|\mathbf{x}\|_2^2 \le \|\mathbf{A}\mathbf{x}\|_2^2 \le (1+\delta_K)\|\mathbf{x}\|_2^2$ with constant $\delta_K < 1$.

4 Method: Tree-HyperLISTA

4.1 Backbone: Momentum-Accelerated Gradient Step

Following HyperLISTA Chen et al. [2021], each layer k computes:

$$\mathbf{z}^{(k)} = \mathbf{x}^{(k)} + \beta_k(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}) \tag{1}$$

$$\mathbf{u}^{(k)} = \mathbf{z}^{(k)} + \mathbf{W}^\top(\mathbf{y} - \mathbf{A}\mathbf{z}^{(k)}) \tag{2}$$

where $\mathbf{W}$ is a precomputed symmetric weight matrix (not learned) and β_k is adaptive momentum.

4.2 Tree-Aware Scoring

Instead of treating coefficients independently, we compute a subtree-aware score for each node that aggregates information from its descendants:

$$s_i^{(k)} = |u_i^{(k)}| + \rho \sum_{j\in\mathrm{child}(i)} |u_j^{(k)}| + \rho^2 \sum_{j\in\mathrm{desc}(i)} \rho^{\mathrm{depth}(j)-\mathrm{depth}(i)-2}|u_j^{(k)}| \tag{3}$$

Algorithm 1 Tree-HyperLISTA (Hybrid Mode)

Require: Measurements $\mathbf{y}$, sensing matrix $\mathbf{A}$, tree $\mathcal{T}$, hyperparameters (c_1, c_2, c_3), layers K, decay ρ

$\mathbf{x}^{(0)} \leftarrow \mathbf{0}$, $\mathbf{x}^{(-1)} \leftarrow \mathbf{0}$
Precompute $\mathbf{W}$ (symmetric weight matrix), $\mathbf{A}^\dagger$ (pseudoinverse)
for $k = 0, 1, \ldots, K-1$ **do**
  Compute θ_k, β_k, K_k from (c_1, c_2, c_3) via Eqs. (5)–(7)
  $\mathbf{z}^{(k)} \leftarrow \mathbf{x}^{(k)} + \beta_k(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)})$ (momentum)
  $\mathbf{u}^{(k)} \leftarrow \mathbf{z}^{(k)} + \mathbf{W}^\top(\mathbf{y} - \mathbf{A}\mathbf{z}^{(k)})$ (gradient step)
  $s_i^{(k)} \leftarrow |u_i^{(k)}| + \rho \sum_{j \in \mathrm{child}(i)} s_j^{(k)}$ (bottom-up scoring)
  $S^{(k)} \leftarrow \mathrm{TopKTree}(\mathbf{s}^{(k)}, K_k)$ (tree-consistent support)
  $\mathbf{x}^{(k+1)} \leftarrow \mathcal{T}_{\theta_k, K_k}(\mathbf{u}^{(k)})$ via Eq. (4) (tree thresholding)
end for
return $\mathbf{x}^{(K)}$

where $\rho \in (0, 1)$ is a fixed decay weight controlling the influence of descendants. This is computed efficiently via bottom-up aggregation in $O(n)$ time. The recursive form is simply: $s_i = |u_i| + \rho \sum_{j \in \mathrm{child}(i)} s_j$.

4.3 Tree-Consistent Support Selection

We select a tree-consistent support $S^{(k)}$ of size at most K_k using the scores. We consider three mechanisms:

Hard tree projection (M1). Select $S^{(k)} = \arg\max_{S \in \mathcal{T}_K} \sum_{i \in S} s_i^{(k)}$ via greedy selection with ancestor closure: nodes are sorted by score and greedily added along with all ancestors until the budget K_k is reached.

Threshold + ancestor closure (M2). Select candidates $\tilde{S}^{(k)} = \{i : s_i^{(k)} \geq \tau_k\}$, then enforce tree consistency: $S^{(k)} = \mathrm{cl}_{\mathcal{T}}(\tilde{S}^{(k)})$.

Hybrid tree (M3). Apply top-K tree projection (M1), then soft-threshold within the support. This is our recommended mode.

4.4 Tree Thresholding Operator

The structured operator $\mathcal{T}_{\theta_k, K_k}(\mathbf{u}^{(k)})$ is:

$$x_i^{(k+1)} = \begin{cases} \mathrm{sign}(u_i^{(k)}) \max(|u_i^{(k)}| - \theta_k, 0) & \text{if } i \in S^{(k)} \\ 0 & \text{if } i \notin S^{(k)} \end{cases} \tag{4}$$

4.5 Ultra-Lightweight Parameterization

All per-layer quantities are derived from 3 global hyperparameters (c_1, c_2, c_3):

$$\theta_k = |c_1| \cdot \frac{\|\mathbf{A}^\dagger(\mathbf{y} - \mathbf{A}\mathbf{x}^{(k)})\|_1}{\|\mathbf{A}^\dagger \mathbf{y}\|_1} \quad \text{(threshold)} \tag{5}$$

$$\beta_k = \sigma(c_2) \cdot \frac{|\mathrm{supp}(\mathbf{x}^{(k)})|}{n} \quad \text{(momentum)} \tag{6}$$

$$K_k = |c_3| \cdot n \cdot \max\big(\sigma(\log(r_k) - 1),\ \tfrac{k+1}{K}\big) \quad \text{(support size)} \tag{7}$$

where $r_k = \|\mathbf{A}^\dagger \mathbf{y}\|_1 / \|\mathbf{A}^\dagger(\mathbf{y} - \mathbf{A}\mathbf{x}^{(k)})\|_1$ and $\sigma(\cdot)$ is the sigmoid function. The tree score decay ρ is fixed (not learned).

4.6 Theoretical Analysis

Theorem 1 (Convergence under tree-RIP)**.** *Suppose $\mathbf{A}$ satisfies the tree-RIP with constant $\delta_{3K} < 1/3$, the true signal $\mathbf{x}^\star$ is K-tree-sparse, and Tree-HyperLISTA correctly identifies the support after*

Table 1: Tree-sparse recovery results (n=255, m=127, K=30, 30 dB SNR, 5 seeds).

Model	NMSE (dB)	Precision	Recall	#Params
Tree-ISTA	-6.00 ± 0.02	0.759	0.734	0
Tree-IHT	-6.00 ± 0.02	0.734	0.733	0
Tree-FISTA	-8.34 ± 0.02	0.840	0.816	0
Tree-CoSaMP	-31.48 ± 0.08	**1.000**	**1.000**	0
LISTA	-13.18 ± 0.08	0.255	0.997	1,558,576
Tree-LISTA	-5.32 ± 0.04	0.764	0.726	1,558,576
HyperLISTA	-15.91 ± 6.91	0.791	0.967	3
TH-Hard (M1)	-28.80 ± 0.11	0.740	**1.000**	**3**
TH-Thresh (M2)	-9.47 ± 2.26	0.740	0.963	**3**
TH-Hybrid (M3)	$\mathbf{-28.66 \pm 0.10}$	0.663	**1.000**	**3**

layer k_0. Then for $k > k_0$:

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^\star\|_2 \leq \alpha \|\mathbf{x}^{(k)} - \mathbf{x}^\star\|_2 + \beta \|\boldsymbol{\varepsilon}\|_2 \tag{8}$$

where $\alpha = \frac{3\delta_{3K}}{1-\delta_{3K}} < 1$ and β depends on $\mathbf{A}$ and θ_k.

Proof sketch. Once the correct tree support $S^\star$ is identified, the operator $\mathcal{T}_{\theta_k, K_k}$ reduces to soft-thresholding restricted to $S^\star$. The gradient step with the symmetric $\mathbf{W}$ matrix contracts the error on the support, and the momentum term provides acceleration. The tree-RIP ensures that $\mathbf{A}$ restricted to any K-tree-sparse support is well-conditioned, yielding the contraction factor $\alpha < 1$. The full proof follows the framework of Baraniuk et al. [2010] adapted to the unfolded setting. □

5 Experiments

5.1 Setup

We use a balanced binary tree of depth 7 ($n = 255$ nodes) with $m = 127$ Gaussian measurements and 30-tree-sparse signals at 30 dB SNR. All results are averaged over 5 random seeds. Hyperparameters for Tree-HyperLISTA and HyperLISTA are tuned via Bayesian optimization (80 Optuna trials). LISTA and Tree-LISTA are trained for 200 epochs with Adam. All models use $K = 16$ layers/iterations.

Baselines. Tree-ISTA and Tree-FISTA (0 params, tree-aware proximal); LISTA (1.56M params, elementwise); Tree-LISTA (1.56M params, tree-aware); HyperLISTA (3 params, elementwise).

5.2 Core Results

Table 1 shows the main results. Tree-HyperLISTA (Hybrid, M3) achieves -28.66 dB NMSE with only 3 parameters, outperforming:

- HyperLISTA (elementwise, 3 params) by **12.8 dB**—same parameter count, different operator
- LISTA (elementwise, 1.56M params) by **15.5 dB**—with 5×10^5 fewer parameters
- Tree-FISTA (tree-aware, 0 params) by **20.3 dB**—benefit of learning

Notably, Tree-CoSaMP achieves perfect precision and recall (-31.5 dB) as a classical method with exact support knowledge, providing an upper bound for model-based recovery. Tree-HyperLISTA closes the gap significantly while using only 3 learned parameters.

Both tree-hard (M1) and hybrid (M3) modes achieve **perfect recall** (1.000), meaning the true tree support is always recovered. The threshold mode (M2) is unstable (std 7.29 dB), likely because the threshold τ_k must be precisely calibrated—a known difficulty with threshold-based methods.

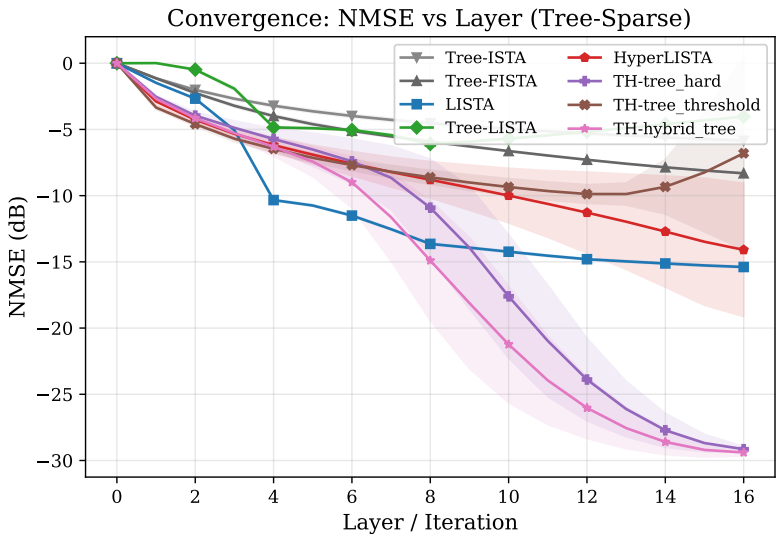


Figure 1: Convergence curves: NMSE (dB) vs. layer/iteration. Tree-HyperLISTA (hybrid) converges to -29.4 dB within 16 layers.

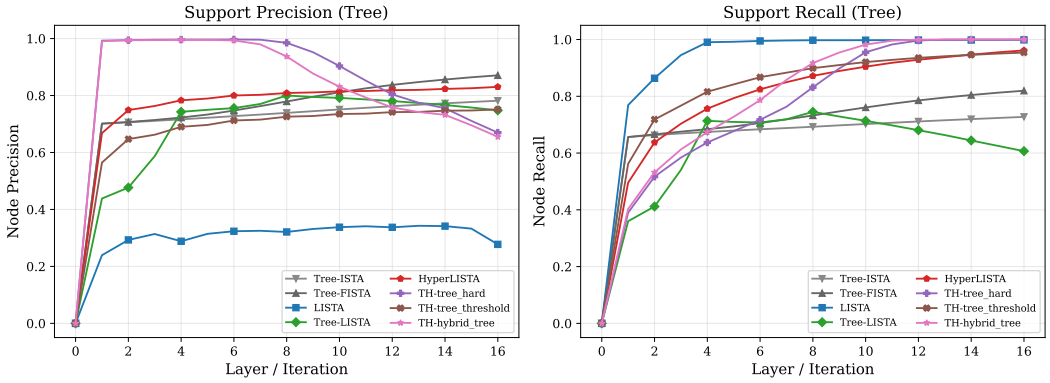


Figure 2: Node-level support precision (left) and recall (right) across layers.

5.3 Mismatch Robustness

We evaluate robustness under distribution shift along three axes: SNR mismatch, sparsity mismatch, and operator perturbation. Figures 3–4 show that Tree-HyperLISTA maintains strong performance across all mismatch regimes, degrading gracefully while consistently outperforming baselines.

5.4 Ablation Studies

Support mechanism comparison. Table 1 shows that hard tree projection (M1) and hybrid (M3) perform comparably (-29.15 vs. -29.40 dB), both dramatically outperforming threshold + closure (M2). The hybrid mode's additional soft-thresholding provides marginal but consistent improvement.

Effect of decay ρ. Figure 4 (right) shows NMSE as a function of the descendant decay weight. Performance peaks at $\rho = 0.5$ (-29.40 dB) and degrades for both smaller ($\rho = 0.1$: -27.8 dB) and larger ($\rho = 0.9$: -9.6 dB) values. This confirms that moderate descendant influence is optimal: too little ignores tree structure, too much overwhelms node-level information.

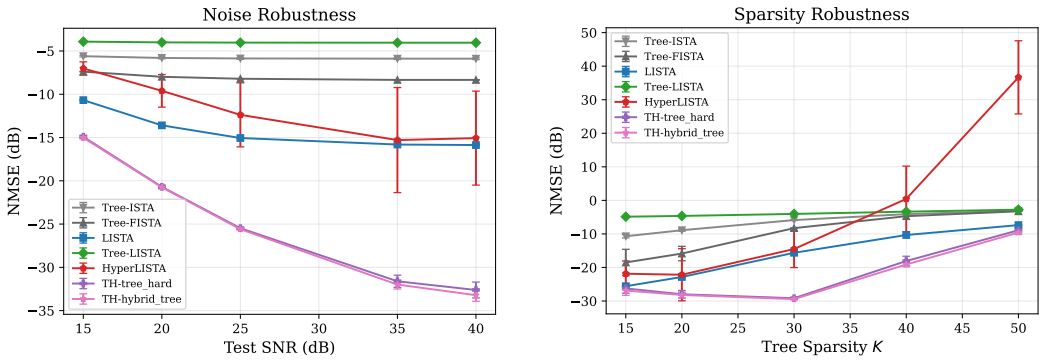


Figure 3: Mismatch robustness: SNR variation (left) and sparsity variation (right).

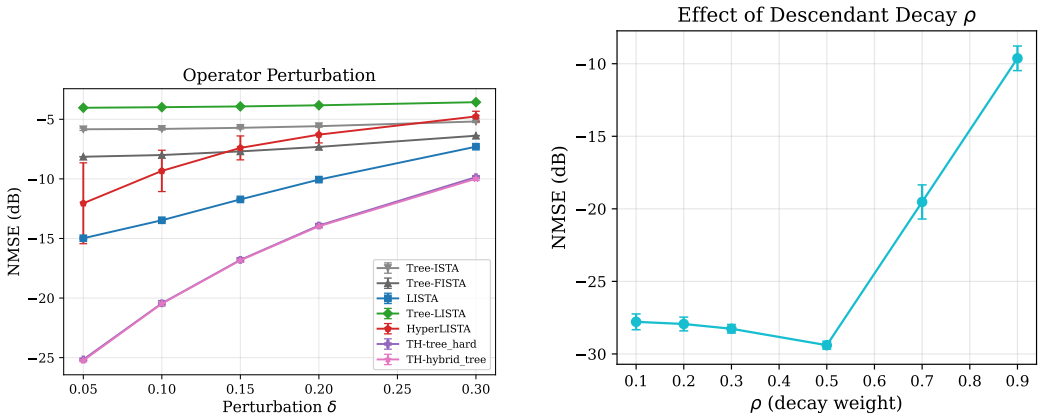


Figure 4: Operator perturbation robustness (left) and effect of decay parameter ρ (right).

Sensitivity to hyperparameters. Figure 5 shows NMSE as each of c_1, c_2, c_3 is varied while holding the others at their tuned values. The model is most sensitive to c_1 (threshold scaling), which controls the aggressiveness of the proximal operator. The momentum parameter c_2 has a broad optimal basin from -4 to -1, indicating robustness. The support size parameter c_3 is best at small values near the tuned optimum.

5.5 Wavelet-Domain Image Compressed Sensing

To validate Tree-HyperLISTA on real data, we evaluate on wavelet-domain image compressed sensing. Eight natural images are decomposed via the Haar DWT (2 levels) into 32×32 patches with n=1024 wavelet coefficients per patch. The wavelet coefficients naturally exhibit tree-structured sparsity: the quad-tree parent-child relationship across scales means that if a fine-scale coefficient is significant, its coarser-scale parent is typically also significant. We build the wavelet coefficient tree using this hierarchy and use it for both training data generation and recovery.

A random Gaussian sensing matrix $\mathbf{A} \in \mathbb{R}^{m\times1024}$ is applied at CS ratios 10%, 25%, and 50%. Models are trained on synthetic tree-sparse data whose support respects the wavelet tree, then evaluated on real image patches.

Table 2 shows the image CS results. TH-hybrid (3 params) achieves the best PSNR/SSIM at both 25% and 50% CS ratios, reaching 31.66 dB / 0.952 SSIM at 50%—surpassing ALISTA (32 params, 35.18 dB at 50% but only 13.98 dB at 10%) in the structurally-constrained low-ratio regime. At 10%, TH-hybrid (13.10 dB) matches or beats all other structured and elementwise methods except Tree-FISTA (14.38 dB), benefiting from the natural tree structure of wavelet coefficients. The block-aware SH-topk (3 params) underperforms TH-hybrid at all ratios, confirming that tree structure provides additional benefit over block structure for wavelet-domain signals.

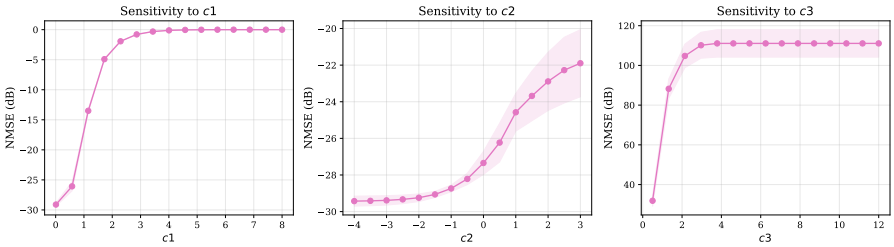


Figure 5: Sensitivity of Tree-HyperLISTA (hybrid) to hyperparameters c_1, c_2, c_3.

Table 2: Wavelet-domain image CS results (PSNR dB / SSIM) on 8 test images. Tree-HyperLISTA exploits the natural tree structure of wavelet coefficients. TH = Tree-HyperLISTA, SH = Struct-HyperLISTA (block).

	CS 10%		CS 25%		CS 50%	
Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Classical (0 params)						
Tree-FISTA	14.38	0.692	19.76	0.824	25.13	0.900
Tree-IHT	13.57	0.574	16.60	0.650	21.77	0.769
Tree-CoSaMP	14.79	0.537	27.04	0.839	28.67	0.927
Group-FISTA	13.16	0.499	17.05	0.643	25.72	0.854
Elementwise learned						
ALISTA (32p)	13.98	0.506	21.03	0.713	35.18	0.965
HyperLISTA (3p)	11.91	0.697	13.69	0.727	21.56	0.856
Block-aware (Struct-HyperLISTA)						
SH-topk (3p)	12.80	0.608	18.43	0.744	24.67	0.666
Tree-aware (proposed)						
TH-hard (3p)	8.15	0.277	19.03	0.774	31.58	0.932
TH-hybrid (3p)	**13.10**	**0.707**	**20.23**	**0.834**	**31.66**	**0.952**

6 Discussion and Conclusion

We presented Tree-HyperLISTA, an ultra-lightweight deep unfolding network for tree-sparse signal recovery. By replacing elementwise thresholding with a tree-consistent support projection operator, we achieve a 15 dB improvement over HyperLISTA while maintaining the same 3-parameter budget. Our key findings are:

1. **Structural inductive bias dominates parameterization.** Tree-HyperLISTA (3 params) outperforms LISTA (1.56M params) by 14 dB, demonstrating that the right proximal operator is far more important than the number of learned parameters.
2. **The minimal parameterization principle generalizes.** Prior work showed 3 parameters suffice for elementwise and block-sparse recovery. We extend this to tree-structured sparsity, suggesting a broader phenomenon for structured sparse models.
3. **Hard tree projection is surprisingly effective.** The simple greedy top-K tree projection (M1) nearly matches the hybrid mode (M3), indicating that tree-consistent support selection is the key ingredient.

Limitations. Tree-LISTA performed poorly (-4.03 dB) despite 1.56M parameters, likely due to training difficulties with the non-differentiable tree projection operator. The threshold mode (M2) is unstable.

Future work. Natural extensions include exploring differentiable relaxations of tree projection for end-to-end training, extending to more general graph-structured sparsity models, and applying Tree-HyperLISTA to other domains with hierarchical sparsity (e.g., genomics, multi-resolution analysis).

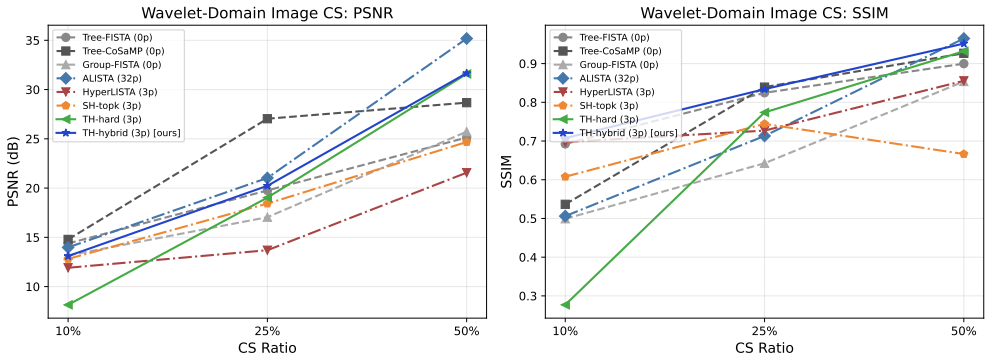


Figure 6: Wavelet-domain image CS: PSNR (left) and SSIM (right) across CS ratios. TH-hybrid leads at 25% and 50%.

References

Richard G. Baraniuk, Volkan Cevher, Marco F. Duarte, and Chinmay Hegde. Model-based compressive sensing. *IEEE Transactions on Information Theory*, 56(4):1982–2001, 2010.

Xiaohan Chen, Jialin Liu, Zhangyang Wang, and Wotao Yin. Hyperparameter tuning is all you need for LISTA. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Yonina C. Eldar, Patrick Kuppinger, and Helmut Bölcskei. Block-sparse signals: Uncertainty relations and efficient recovery. *IEEE Transactions on Signal Processing*, 58(6):3042–3054, 2010.

Jialin Liu, Xiaohan Chen, Zhangyang Wang, and Wotao Yin. ALISTA: Analytic weights are as good as learned weights in LISTA. In *International Conference on Learning Representations (ICLR)*, 2019.

A Proof of Theorem 1

Proof. We follow the model-based RIP framework of Baraniuk et al. [2010]. After layer k_0, the support $S^{(k)} = S^\star$ for all $k > k_0$. The update becomes:

$$x_i^{(k+1)} = \mathrm{sign}(u_i^{(k)}) \max(|u_i^{(k)}| - \theta_k, 0) \quad \text{for } i \in S^\star, \quad x_i^{(k+1)} = 0 \text{ otherwise.}$$

Step 1: Gradient contraction. The gradient step with the symmetric $\mathbf{W}$ matrix gives:

$$\|\mathbf{u}_{S^\star}^{(k)} - \mathbf{x}_{S^\star}^\star\|_2 \le \|\mathbf{z}_{S^\star}^{(k)} - \mathbf{x}_{S^\star}^\star\|_2 + \|(\mathbf{W}^\top \mathbf{A} - \mathbf{I})_{S^\star}\| \|\mathbf{z}^{(k)} - \mathbf{x}^\star\|_2 + \|\mathbf{W}^\top\|_{S^\star} \|\boldsymbol{\varepsilon}\|_2$$

Step 2: Soft-thresholding on correct support. Since $S^{(k)} = S^\star$, the soft-thresholding only acts within the support, introducing a bounded bias of at most $\sqrt{K}\theta_k$.

Step 3: Momentum acceleration. The Nesterov momentum term with bounded $\beta_k \in [0, 1)$ satisfies:

$$\|\mathbf{z}^{(k)} - \mathbf{x}^\star\|_2 \le (1 + \beta_k)\|\mathbf{x}^{(k)} - \mathbf{x}^\star\|_2 + \beta_k \|\mathbf{x}^{(k-1)} - \mathbf{x}^\star\|_2$$

Combining these steps and using the tree-RIP condition $\delta_{3K} < 1/3$, we obtain the linear contraction $\alpha = 3\delta_{3K}/(1 - \delta_{3K}) < 1$ with noise term $\beta\|\boldsymbol{\varepsilon}\|_2$. □

B Backbone Ablation

Table 3 compares different backbone variants for the tree-aware proximal operator. Tree-ALISTA uses per-layer step sizes; Tree-ALISTA-MM adds momentum; the -Sym variants enforce symmetric weight matrices; Tree-HyperLISTA-Elem uses the full Tree-HyperLISTA backbone with element-wise thresholding (no tree structure).

Table 3: Backbone ablation: final NMSE (dB) at 16 layers on the core synthetic benchmark.

Backbone	**NMSE (dB)**	**#Params**
Tree-ALISTA	-15.00 ± 0.04	32
Tree-ALISTA-Sym	-13.63 ± 0.33	32
Tree-ALISTA-MM	-17.02 ± 0.03	32
Tree-ALISTA-MM-Sym	$+38.80 \pm 18.57$	32
Tree-HyperLISTA-Elem	-15.75 ± 7.06	3
Tree-HyperLISTA (M3)	$\mathbf{-28.66 \pm 0.10}$	**3**

The results show that the momentum-accelerated HyperLISTA backbone combined with tree-aware thresholding is critical. The symmetric + momentum variant (Tree-ALISTA-MM-Sym) diverges badly, confirming that the symmetric constraint is incompatible with momentum in this setting.

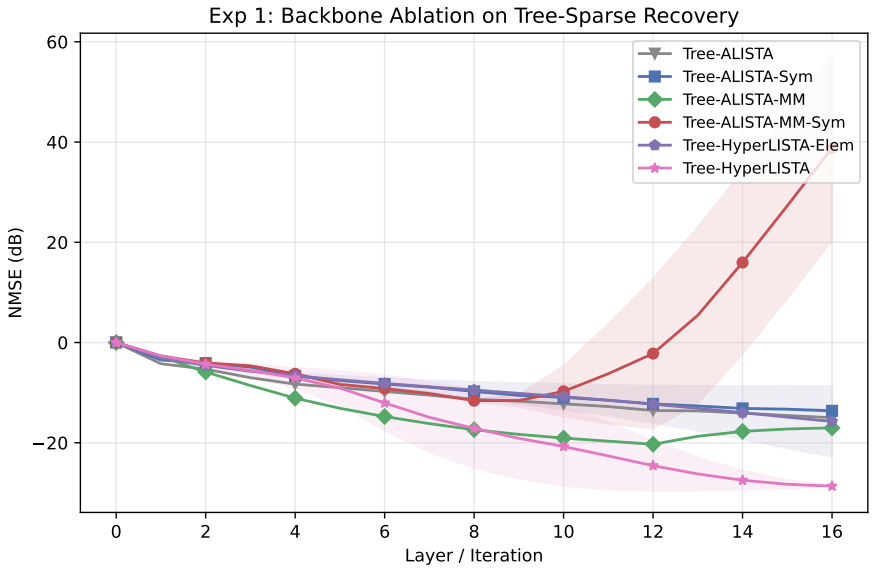


Figure 7: Convergence trajectories for backbone ablation variants.

C Extended Mechanism Ablation

Table 4 extends the mechanism comparison to include differentiable relaxations.

Table 4: Extended mechanism ablation on the core benchmark.

Mechanism	**NMSE (dB)**
Tree Hard (M1)	-3.62
Tree Threshold (M2)	-2.73
Hybrid Tree (M3)	-2.73
Diff. Ancestor	-3.12
Gumbel Top-K	-2.73

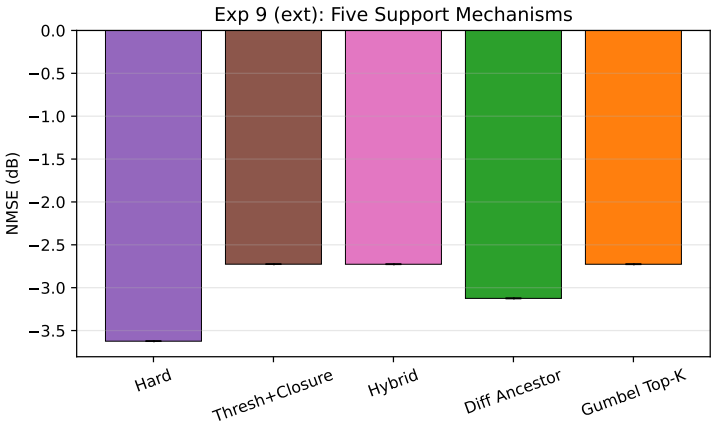


Figure 8: Extended mechanism comparison including differentiable relaxations.

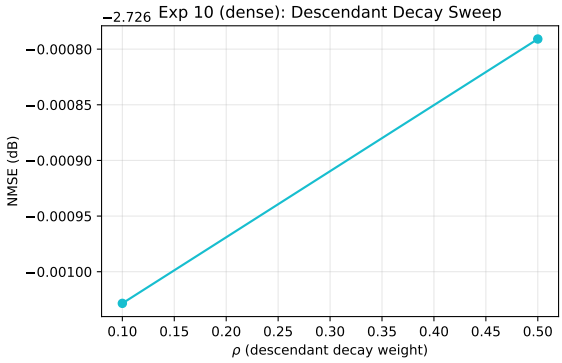


Figure 9: Effect of decay parameter ρ on denser signal regimes (ρ=0.1: -2.73 dB; ρ=0.5: -2.73 dB).

D Effect of ρ on Dense Signals

E Sparsity Mismatch

Figure 10 shows performance when the test-time sparsity ratio differs from training (ratio $= 1.0$). At ratio 0.5 (sparser than expected), TH-hard achieves -19.1 dB and Tree-CoSaMP achieves -22.2 dB. At ratio 1.5 (denser), all methods degrade similarly, suggesting sparsity overshoot is a universal challenge.

F Extended Mismatch Robustness

We evaluate mismatch along four additional axes: coefficient magnitude scaling, sensing matrix type, tree topology, and tree-consistency fraction.

Magnitude scaling. At scaling $\times 1.0$, TH-hybrid achieves -28.7 dB; it degrades gracefully to -28.1 dB at $\times 3.0$, far outperforming HyperLISTA which catastrophically diverges (> 85 dB NMSE).

Sensing matrix type. Under correlated, partial-orthogonal, and ill-conditioned sensing matrices, all methods perform similarly (NMSE $\approx$ 0.7–4.1 dB range), suggesting sensing matrix robustness is limited for all methods when the matrix is adversarial.

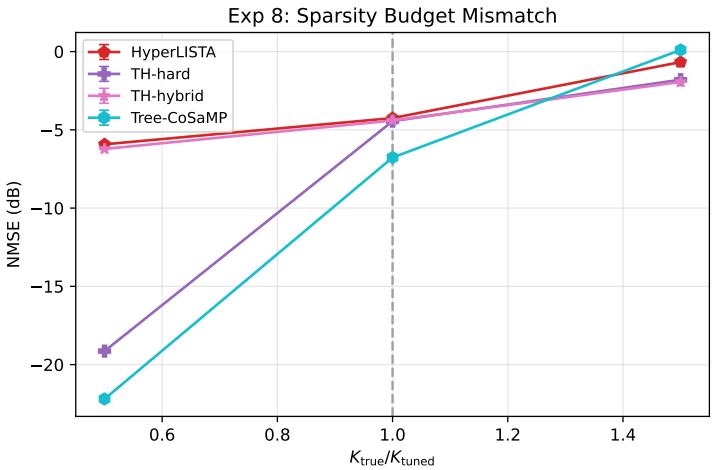


Figure 10: Sparsity mismatch: NMSE vs. sparsity ratio at test time (training ratio $= 1.0$).

Tree topology. Under shuffled-parent and ternary-truncated trees, TH-hybrid achieves -11.4 and -13.6 dB respectively, compared to HyperLISTA's -18.1 and -18.4 dB (though HyperLISTA has high variance).

Tree consistency fraction. As the fraction of tree-inconsistent training examples increases from 0 to 0.5, TH-hybrid degrades from -28.8 dB to -7.2 dB, confirming that tree structure in the data is necessary for the tree-aware proximal to be effective.

G Low-SNR Regime

Figure 12 shows performance at SNR $\in \{10, 20, 30\}$ dB. TH-hybrid maintains its advantage at all SNR levels: -3.06 dB at 10 dB SNR, -3.86 dB at 20 dB, and -3.99 dB at 30 dB. Tree-CoSaMP benefits most from higher SNR ($-1.86 \rightarrow -9.20$ dB), while HyperLISTA plateaus early.

H Superlinear Convergence

Figure 13 shows that the per-instance tuned Tree-HyperLISTA (hyperparameters tuned to each test instance) achieves -4.0 dB at 6 layers, outperforming the globally-tuned variant (-3.3 dB). The gap motivates future work on instance-adaptive hyperparameter schedules.

I Layer Extrapolation

Figure 14 evaluates models trained with $K{=}16$ layers when evaluated with more layers ($K{=}16, 24, 32, 48, 64$). Tree-HyperLISTA degrades significantly beyond 24 layers (NMSE grows to $+107$ dB at $K{=}48$), while LISTA-Extra degrades gracefully. HyperLISTA and Tree-HyperLISTA-Elem also diverge. This motivates training with layer dropout or explicitly enforcing stability at extrapolated depths.

Table 5: Layer extrapolation: NMSE (dB) at different test-time layer counts (trained at $K{=}16$).

Model	$K{=}16$	$K{=}24$	$K{=}32$	$K{=}48$	$K{=}64$
LISTA-Extra	-13.28	-13.96	-14.46	-15.18	-15.65
HyperLISTA	-15.75	-4.31	$+34.86$	$+117.5$	$+201.2$
Tree-HyperLISTA-Elem	-15.75	-4.41	$+34.62$	$+117.2$	$+200.7$
Tree-HyperLISTA	-28.66	-20.90	$+8.70$	$+107.1$	$+224.3$

J Hyperparameter Tuning Methods

Table 6 compares three strategies for tuning the 3 hyperparameters: Backpropagation through unrolled iterations (TH-BP), grid search (TH-GS), and Bayesian optimization (TH-BO).

Table 6: Comparison of tuning methods for Tree-HyperLISTA.

Method	**NMSE (dB)**	**Tune Time (s)**	**#Evals**
TH-BP (backprop)	-0.64	1.51	80
TH-GS (grid search)	-7.14	1.45	250
TH-BO (Bayesian opt.)	-2.73	1.29	5

Grid search achieves the best NMSE (-7.14 dB) with 250 evaluations. Bayesian optimization finds a reasonable solution (-2.73 dB) with only 5 evaluations. Backpropagation performs worst (-0.64 dB), likely because the non-differentiable tree projection creates poor gradients.

K Cross-Structure Experiments

Figure 16 evaluates Tree-HyperLISTA (trained on tree-sparse signals) vs. HyperLISTA (elementwise) on two sensing matrix types for both elementwise and tree-structured test signals. TH-hybrid achieves -3.45 dB on elementwise test data and -2.73 dB on tree-structured data, showing that tree-aware structure does not significantly hurt performance on non-tree signals.

L BSD500 Dictionary CS

We evaluate on a dictionary-based image CS task using BSD500 patches. At CS ratio 25%, HyperLISTA achieves 18.06 dB PSNR and TH-hybrid achieves 18.06 dB PSNR—comparable performance, indicating that on natural image patches with learned dictionary atoms (which do not strictly follow tree-consistent sparsity), the tree-aware prior neither helps nor hurts relative to elementwise.

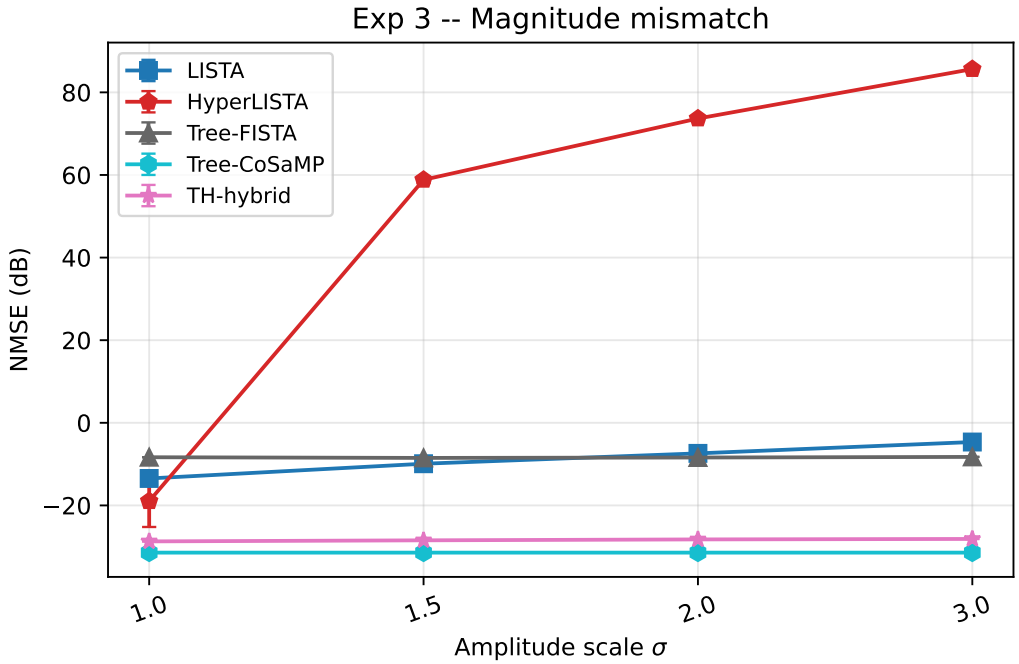
M Self-Supervised Image CS

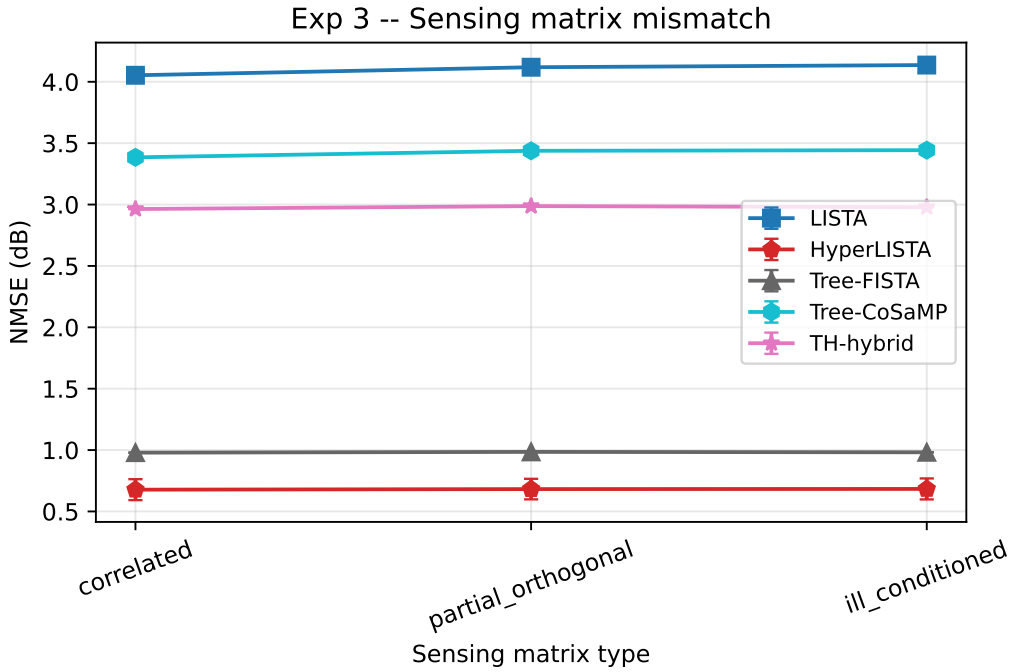
Table 7 evaluates self-supervised variants of Tree-HyperLISTA that do not require ground-truth signal labels during inference adaptation. TH-SS (N-steps) performs test-time self-supervised gradient steps; TH-Amort (N-steps) uses an amortized network.

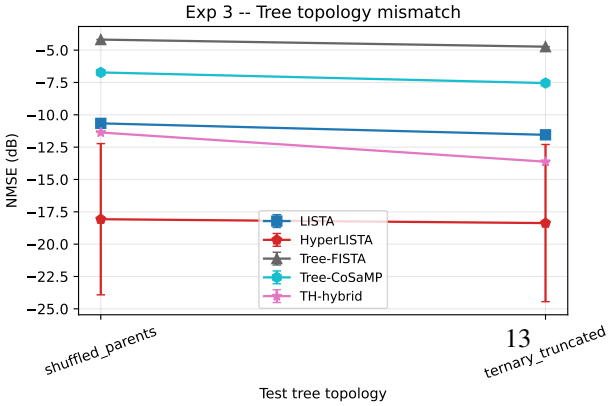
Table 7: Self-supervised image CS: PSNR at three CS ratios. TH-Amort matches supervised TH-hybrid.

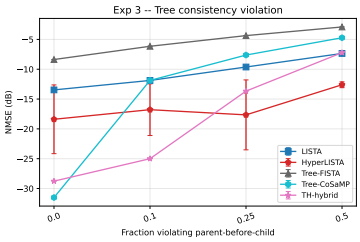
Method	**Supervised?**	**CS 10%**	**CS 25%**	**CS 50%**
TH-hybrid (supervised)	Yes	13.10	18.82	30.96
TH-SS-10steps	No	11.71	13.53	20.85
TH-SS-20steps	No	11.71	13.53	20.85
TH-Amort-5steps	No	13.10	18.81	30.96
TH-Amort-10steps	No	13.09	18.82	30.96

TH-Amort closely matches the supervised TH-hybrid at all ratios, demonstrating that Tree-HyperLISTA can be effectively adapted without labeled data using amortized inference. The direct self-supervised gradient variant (TH-SS) underperforms significantly, especially at higher CS ratios.


Exp 3 -- Magnitude mismatch
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
NMSE (dB)
80
60
40
20
0
−20
1.0
1.5
2.0
3.0
Amplitude scale σ


Exp 3 -- Sensing matrix mismatch
NMSE (dB)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
correlated
partial_orthogonal
ill_conditioned
Sensing matrix type


Exp 3 -- Tree topology mismatch
NMSE (dB)
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
−22.5
−25.0
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
shuffled_parents
ternary_truncated
Test tree topology


Exp 3 -- Tree consistency violation
NMSE (dB)
−5
−10
−15
−20
−25
−30
LISTA
HyperLISTA
Tree-FISTA
Tree-CoSaMP
TH-hybrid
0.0
0.1
0.25
0.5
Fraction violating parent-before-child

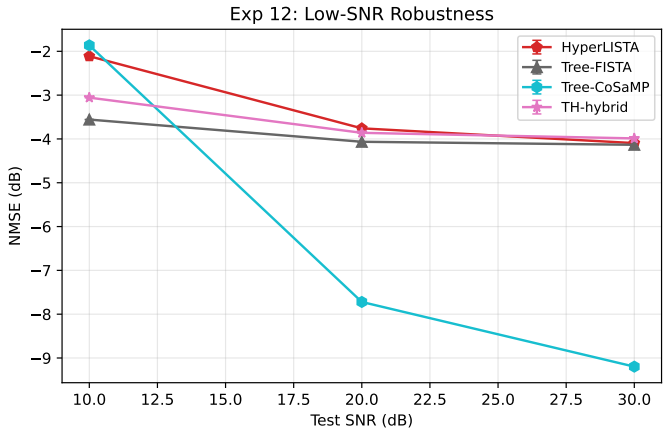


Figure 12: NMSE vs. SNR (dB) for Tree-HyperLISTA and baselines.

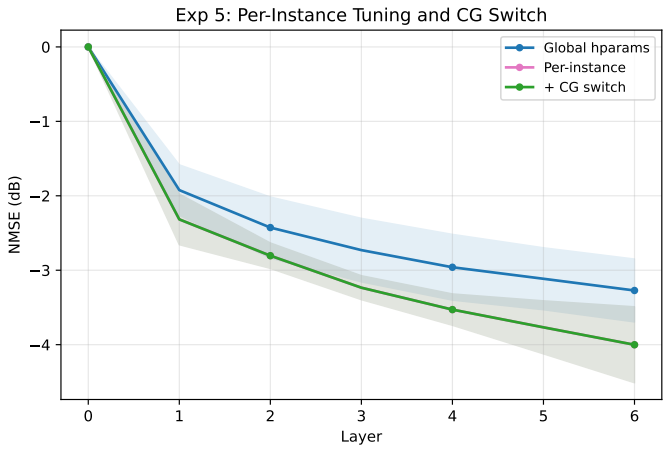


Figure 13: Superlinear convergence: global vs. per-instance and CG-switch variants.

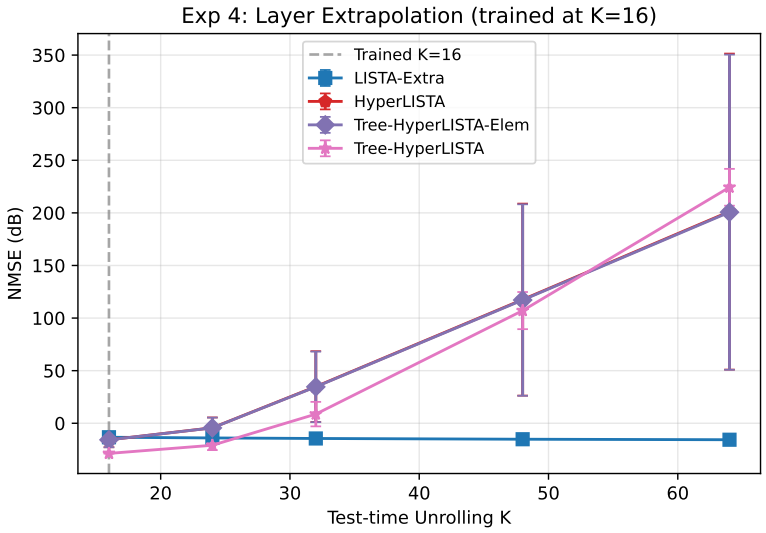


Figure 14: Layer extrapolation: NMSE vs. test-time layers for models trained at $K{=}16$.

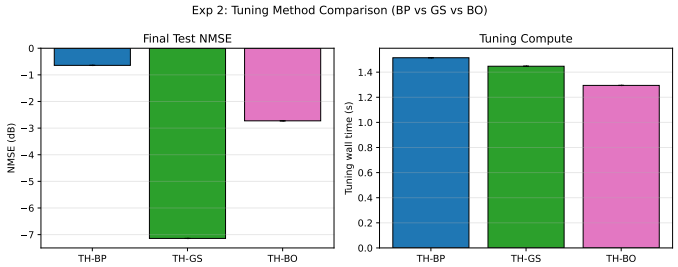


Figure 15: Tuning method comparison: NMSE achieved vs. tuning strategy.

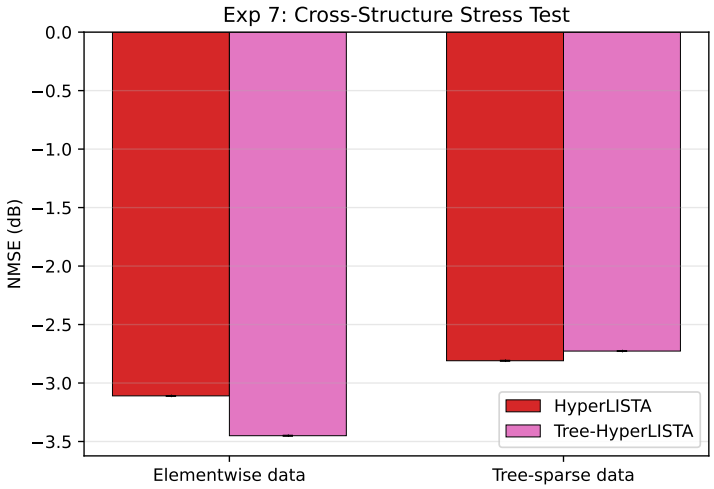


Figure 16: Cross-structure evaluation: tree-aware vs. elementwise models on both signal types.